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| **Fiscal Year:** | FY 2025 | **Task Last Updated:** | FY 09/16/2024 |
| **PI Name:** | Calaway, Kimesha | | |
| **Project Title:** | rHealth One Flight Demonstration | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | Yes |
| **Human Research Program Elements:** | (1) **ExMC**:Exploration Medical Capabilities | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | calawayk@zin-tech.com | **Fax:** | FY |
| **PI Organization Type:** | INDUSTRY | **Phone:** | |
| **Organization Name:** | Zin Technologies, Inc. | | |
| **PI Address 1:** | GRC-MS00 | | |
| **PI Address 2:** | 6745 Engle Road | | |
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| **City:** | Cleveland | **State:** | OH |
| **Zip Code:** | 44130-7994 | **Congressional District:** | 16 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | Directed Research |
| **Start Date:** | 11/18/2019 | **End Date:** | 07/31/2022 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Lemery, Jay | **Contact Phone:** | |
| **Contact Email:** | jay.lemery@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | August 2021 (Ed. note): Kimesha Calaway is Principal Investigator (PI) as of January 2021 and project is in her name for the entire project period; Gail Perusek is CoInvestigator (CoI). Original PI was John Zoldak, who left the project in mid-2020; Cody Farinacci took over for a few months; Calaway has been involved from beginning. Beth Lewandowski was original CoI, no longer on the project. | | |
| **COI Name (Institution):** | Perusek, Gail M.S. ( NASA Glenn Research Center ) | | |
| **Grant/Contract No.:** | Directed Research | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | For current spaceflight missions, NASA has the ability to return blood and urine samples to the ground for testing and evaluation. For long duration space missions, it will not be possible for ground testing of these flight samples. There is a need to evaluate if Commercial-Off-the-Shelf (COTS) medical devices, which can assay biologic fluids, function correctly in microgravity. The rHEALTH ONE platform will be certified for flight using the JPD 7120.9 Experimental Flight Hardware (Class 1-E) Development Process. Vendor will perform minor modifications to the device to conform to flight certification requirements. The flight device will demonstrate flow cytometry – particle counts, intensity, measurements of immunoassay markers with control samples that are representative of blood cells. Those flight measurements will be compared to ground measurements taken utilizing the same batched control samples. The on-orbit demonstration will show that the rHEALTH ONE operates nominally in the microgravity environment. |
| **Rationale for HRP Directed Research:** | This research is directed because it contains highly constrained research. In Planning, Programming, Budgeting, and Execution (PPBE) 20, Exploration Medical Capability (ExMC) Human Research Program (HRP) element planned an ISS blood analysis technology demonstration in FY20. The rHealth ONE is the implementation of that scope of work. Task complies with the intent of Technology Development activities. Task TRL (Technology Readiness Level) is 5/6. |
| **Research Impact/Earth Benefits:** | Monitoring crew health on deep-space exploration missions presents unique challenges, including limited space for medical devices and the inability to return samples to Earth for analysis. The rHEALTH demonstration tested using a modified, commercial off-the-shelf device to diagnose certain medical conditions. The device utilizes flow cytometry, a method using lasers to sort and identify cells, and can analyze cell count and cell characteristics; detect microorganisms, biomarkers, and proteins; and diagnose health disorders such as blood cancers. The demonstration verified that the hardware functions in the space environment and included performance evaluation against results from ground testing. This technology also could provide timely, cost-effective, reliable, and convenient diagnostic tests for patients on Earth who lack access to robust health-care infrastructure. |
| **Task Progress:** | The rHealth One was launched to the International Space Station (ISS) on February 19, 2022.<br>The rHealth One demonstrations on the ISS were completed on May 13, 2022, and May 16, 2022, where microliter drops of control samples were analyzed with the rHEALTH One.<br><br>The data from the analyzer was downlinked from the ISS and post processing was completed during the summer of 2022.<br><br>Final publications were completed by February 2024.<br><br>There is no further work being done on this task. |
| **Bibliography Type:** | Description: (Last Updated: 10/15/2024) |
| **Articles in Peer-reviewed Journals** | Rea DJ, Miller RS, Crucian BE, Valentine RW, Cristoforetti S, Bearg SB, Sipic Z, Cheng J, Yu R, Calaway KM, Eames D. "Single drop cytometry onboard the International Space Station." Nature Communications. 2024 Mar 25;15(1):2634. https://doi.org/10.1038/s41467-024-46483-6 ; PMID: 38528030 ; PMCID: PMC10963801 , Mar-2024 |
| **Articles in Peer-reviewed Journals** | Crucian B, Valentine R, Calaway K, Miller R, Rubins K, Hopkins M, Salas Z, Krieger S, Makedonas G, Nelman-Gonzalez M, McMonigal K, Perusek G, Lehnhardt K, Easter B. "Spaceflight validation of technology for point-of-care monitoring of peripheral blood WBC and differential in astronauts during space missions." Life Sci Space Res. 2021 Nov;31:29-33. https://doi.org/10.1016/j.lssr.2021.07.003 ; PMID: 34689947 , Nov-2021 |
| **Articles in Peer-reviewed Journals** | Lewandowski BE, Schkurko CM, Miller RS, Valentine RW, Calaway KM, Yang JD, Ebert DJ, Sargsyan A, Byrne V, Walton M, Lemery J, Suresh R, Thompson MS, Easter BD, Lehnhardt KR. "Technology modification, development, and demonstrations for future spaceflight medical systems at NASA." Front Space Technol. 2024 Jul 14;5:1384457. https://doi.org/10.3389/frspt.2024.1384457 , Jul-2024 |
| **Significant Media Coverage** | Smith MS (article on rHealth One, a single-drop flow cytometer). "This blood-sampling cytometer is small enough for Mars: The long voyage will require portable ways to monitor astronaut health." IEEE Spectrum, April 18, 2024. https://spectrum.ieee.org/astronaut-health-in-space , Apr-2024 |